

2026 软件体系结构真实考试回忆卷

2026 Software Architecture Real Exam Reconstruction

南京大学软件学院研究生《软件体系结构》Graduate Software Architecture, NJU Software Institute

根据考后题面记录整理Reconstructed from post-exam notes

闭卷英文题面，可用中文作答Closed-book paper in English; Chinese answers were allowed · 100 points

题面按考后记录重构，答案按完整复习课录音、复习课课件和课程 slides 的口径整理；设计题材料原文不完整，因此答案按记录到的实验耗材申领与采购场景保守作答。Questions are reconstructed from post-exam notes. Answers are grounded in the complete review transcript, review slides, and course slides. The design narrative was not fully recorded, so the design answer is conservative and scenario-based.

部分	分值	说明
一、简答题Part I. Short Answers	60	10 题，每题 6 分。10 questions, 6 points each.
二、分析论述题Part II. Analysis	20	2 题，每题 10 分。2 questions, 10 points each.
三、设计题Part III. Design	20	实验耗材申领与采购平台，3 个小问。Experiment consumable procurement and requisition platform, 3 tasks.
总分	100	2026 真实考试回忆整理



答题说明

- 题面和参考答案都支持中英文切换；中文适合背诵，英文用于对照考试题面。Both questions and answers can be switched between Chinese and English. Chinese is optimized for memorization; English is for exam-text alignment.
- 简答题按“概念定义 + 关系/区别 + 例子或权衡”组织；画图题先写画法，再给图或表。For short answers, use definition, relationship/difference, and example/trade-off. For drawing questions, describe how to draw before giving the sketch/table.
- 设计题的事件命名使用过去时业务事实，不写“点击按钮”这类 UI 操作。For the design question, name domain events as past-tense business facts, not UI actions such as clicking a button.
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PART-I

60 pts

一、简答题Part I. Short Answers

每题 6 分，重点是把概念边界、关系和例子写清楚。6 points each; focus on concepts, relationships, and examples.

QI-1

6 PTS • P0 • EASY / 易

说明软件需求、质量属性和架构显著需求（ASR）的区别与关系。Explain the difference and relationship among software requirements, quality attributes, and Architecturally Significant Requirements (ASRs).

QI-2

6 PTS • P1 • MEDIUM / 中

列出 6 个通用架构设计策略，并结合具体例子说明它们如何用于架构设计。List six general architecture design strategies and explain how they are used in architecture design with concrete examples.

QI-3

6 PTS • P0 • EASY / 易

如何建模质量属性场景？用刺激-响应图画出可用性和可修改性的质量属性场景。How do we model quality attribute scenarios? Draw stimulus-response scenarios for availability and modifiability.

QI-4**6 PTS • P0 • MEDIUM / 中**

说明架构模式和战术的区别与关系，并给出两个模式例子及其关联的若干战术。

Explain the difference and relationship between architectural patterns and tactics. Give two concrete pattern examples and several related tactics.

QI-5**6 PTS • P0 • MEDIUM / 中**

给出架构设计主要过程步骤，并说明每一步的输入和输出。Give the major architecture design process steps and the input and output of each step.

QI-6**6 PTS • P0 • MEDIUM / 中**

相较于面向对象设计，分析在领域驱动设计中引入聚合模式的价值。Compared with Object-Oriented Design, analyze the value of introducing the Aggregate pattern in Domain-Driven Design.

QI-7**6 PTS • P0 • EASY / 易**

为什么文档化架构设计时需要不同视图？给出 4 个示例视图的名称和目的。Why are different views needed when documenting architecture design? Give the name and purpose of four example views.

QI-8**6 PTS • P0 • HARD / 难**

给出微服务系统中的 4 个架构元素，并说明它们各自的角色。 **Give four architectural elements of a microservice system and explain the role of each.**

QI-9**6 PTS • P0 • MEDIUM / 中**

说明事件驱动架构风格的基本思想，并给出它在质量属性上的权衡。 **Explain the basic idea of the event-driven architectural style and its trade-offs on quality attributes.**

QI-10**6 PTS • P1 • MEDIUM / 中**

给出企业架构中的 4A，并说明它们之间的关系。 **Give the 4A in enterprise architecture and explain their relationship.**

PART-II

20 pts

二、分析论述题Part II. Analysis Questions

每题 10 分，要写出分类、影响和场景化论证。10 points each; classify, analyze effects, and argue with a system scenario.

QII-
1

10 PTS • PO • MEDIUM / 中

简述 7 类设计决定；说明 **binding time** 有哪些选择，并分析不同绑定时间会影响哪些质量属性。Describe seven categories of design decisions. Give possible choices of binding time and analyze how different binding times affect quality attributes.

QII-
2

10 PTS • PO • HARD / 难

选择一个具体系统场景，论述软件架构从 C/S 演进到分层/三层，再到 SOA 的原因、解决的问题和局限。Choose a concrete system scenario and discuss why the architecture evolves from C/S to layered/three-tier architecture and then to SOA, including context, solved problems, and limitations.

PART-III

20 pts

三、设计题Part III. Design Question

实验耗材申领与采购平台，按 DDD 事件风暴和聚合作答。Experiment consumable procurement and requisition platform; answer with DDD event storming and aggregates.

**20 PTS • P0 • HARD / 难**

实验耗材采购与申领平台设计题：按题面给出的申请、审批、采购、入库等流程，整理领域事件流、策略和聚合。**Design question: For an experiment consumable procurement and requisition platform, organize the domain event flow, policies, and aggregates from the given requisition, approval, procurement, and warehousing process.**